

Jaemin Jeong

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Research Interests

Interested in **polymer adhesion and interfacial design**, particularly how molecular structure and interfacial interactions govern adhesion and performance. My interest grew from observing partial electrode-layer delamination during LCO cathode fabrication. I hope to investigate **electrode binders and functional polymer systems** for energy and mobility applications.

Education

Chungnam National University

Daejeon, Republic of Korea

B.S. Student, Organic Materials Engineering

Feb 2023 – Present

- Transferred internally from Rural and Environmental Engineering to Organic Materials Engineering in Feb 2023.
- 51 major credits completed; cumulative GPA 3.37/4.5; major GPA 3.29/4.5.
- Most recent semester (Spring 2026): **major GPA 3.83/4.5** over 18 credits.

Republic of Korea Air Force

Mandatory military service

2023 – 2025

Project Experience

LCO Cathode Half-Cell Fabrication & Electrochemical Evaluation

Korea Battery Academy

Team of 2; hands-on fabrication and electrochemical evaluation

Jul – Aug 2026

- Formulated and dispersed an LCO/Super P/PVDF (94:3:3) cathode slurry; fabricated electrodes by doctor-blade coating, vacuum drying, roll pressing, and 14 mm punching (active-material loading: **14.7 mg/cm²**).
- Assembled CR2032 half-cells against Li metal in an Ar-filled glovebox and performed Formation, C-rate testing (0.1–2.0C), and initial cycling evaluation over 3.0–4.2 V vs. Li/Li⁺.
- Observed up to **45% variation in initial cell capacity** from electrodes prepared from the same sheet and partial electrode-layer delamination after pressing, motivating an interest in **binder-current-collector interfacial adhesion**.

Battery Materials Characterization

Korea Battery Academy

DSC, TGA, Raman, ICP-OES, GC-MS, and IC/MS; Team of 4

Jul – Aug 2026

- Interpreted polymer and separator thermal behavior using DSC Heat–Cool–Heat analysis (T_g , crystallization, and T_m) and evaluated thermal decomposition behavior using TGA.
- Interpreted Raman D/G/2D bands to assess graphitic carbon structure and relative defect characteristics; compared signal quality under different laser conditions.
- Quantified electrolyte solvents/additives using GC-MS (SIM mode; calibration $R^2 > 0.999$) and quantified Cu using ICP-OES at 324.754 nm (calibration $R^2 = 0.99998$; $RSD = 0.395\%$ across three replicates).

Laboratory Skills

Electrode fabrication: slurry formulation/dispersion, doctor-blade coating, vacuum drying, roll pressing, punching, viscometry, interface/delamination observation

Cell assembly & electrochemistry: Ar-glovebox handling, CR2032 assembly, Formation, C-rate testing, initial cycling, charge–discharge profile interpretation

Materials characterization: DSC, TGA, Raman, ICP-OES, GC-MS, IC/MS

Computing: Python (pandas, matplotlib) for experimental data handling and visualization

Training & Awards

Secondary Battery Materials Analysis & Recycling – 132-hour Program

Korea Battery Academy / KBIA

Certificate No. 2026-00048

Jul – Aug 2026

Daegu Public Data Startup Competition – Award

Daegu Digital Innovation Promotion Agency

Idea & Planning Division; public-data collection and preprocessing

2025

Relevant Coursework

Polymer Synthesis (A0) | Polymer Materials (A0) | Industrial Chemistry I (A+) | Engineering Physics (A0)

Organic Material Properties | Organic Materials | Introduction to Polymer Engineering | Physical Chemistry & Laboratory | Introduction to Materials Engineering | Textile Chemistry | Petrochemical Industry